A Statistical Examination of Factors Influencing Retail Transaction Value

CA 4 – Cross-Module Project Report for Data Science

Module: RESA C9009 – Research Process for Data Analytics

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Abstract

This project analyses a retail transaction dataset containing 12,575 individual purchases and hopes to identify the factors that influence transaction values and to understand spending behavior varies across product types, time periods and payment methods. Each transaction includes information on the product category, price per unit, quantity purchased, total spend, payment method, discount status and variables such as weekday or weekend classification. A complete data preprocessing workflow was used to clean missing values, standardise the categorical labels, generate time based features and to review the outliers to ensure the dataset was suitable for statistical analysis.

Four research questions were explored in this anlaysis. First a one-way ANOVA tested whether average transaction values differ across product categories. Second an independent samples t-test found whether weekday and weekend spending differ. Third a Chi-square test of independence supported by logistic regression found whether payment method usage varies by product category. Finally multiple linear regression models were applied to assess whether transaction value can be predicted using behavioural and categorical variables and whether predictive accuracy improves when including numerical variables such as unit price and quantity.

The results show that product category has a statistically significant effect on transaction value, while overall differences between weekday and weekend spending are not significant. Payment method usage appears consistent across categories. The initial regression model explains very little variance ($R^2 = 0.004$) but an improved model using price and quantity explains 83.3 percent of transaction value variance highlighting these as the main influences of spending. The analysis demonstrates the value of combining preprocessing with appropriate statistical methods to generate meaningful insights in retail analytics.

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1 Introduction and Problem Statement

Retailers use data-driven insights to understand how customers spend, what products they buy and the factors that influence these decisions. Payment technologies are changing and consumer behavior becomes more unpredictable. Transation-level data provides a useful way of identifying patterns that support decision-making in areas like pricing, promotions, inventory management, and payment strategy. Understating these factors that drive transaction value is important as it directly impacts revenue,customer lifetime value and overall financial performance of retail businesses.

The dataset contains 12,575 individual retail transactions that each represent a customer purchase record across a different product categories. For each transaction the dataset includes key attributes like the product category, price per unit, quantity purchased, total spend, payment method, location, and transaction date and additional variables extracted from the date including day of week and weekend status. The dataset supports both behavioral and time based analyses and modelling tasks so that transaction value can be predicted from the characteristics of the purchase.

The aim of this project is to identify which factors meaningfully influence transaction value and to determine whether spending behavior varies across categories, days, and payment methods. The objectives in the research mentioned in the earlier research proposal which emphasised finding how consumer spending differs across contexts and how statistical modelling can reveal these patterns. The analysis uses a combination of descriptive statistics, inferential tests, and predictive modelling techniques.

To address the main objective, four focused research questions are formulated:

- Do average transaction values differ across product categories?
- Is there a significant difference in transaction value between weekday and weekend purchases?
- Is payment method associated with product category?
- Can transaction value be predicted from behavioral and categorical variables, and how does model performance change when numeric variables such as price and quantity are included?

These research questions give the structure of the report and the selection of different approaches. They provide an insight into customer spending patterns to find whether specific relationships exist but also how strongly various factors contribute to explaining transaction value. The purpose of this project is to demonstrate the value of structured statistical analysis for uncovering actionable retail insights.

2 Motivation

Understanding what drives customer spending is important for retailers to be successful in competitive environments. Transaction value directly contributes to revenue and small shifts in the average spend can have meaningful financial implications when we look at a scale of thousands of products.

From a commercial perspective looking into transaction value several key decision-making areas can be identified. Retailers can use this information to optimise pricing strategies, identify categories that generate higher-value purchases and find which product groups could benefit from targeted promotions. Time based patterns in spending like differences between weekday and weekend behaviour can help with inventory planning and promotional timing. Understanding payment method usage like whether customers prefer cash, card, or digital wallets can guide investments in payment infrastructure and promotional incentives for specific methods.

Transaction-level data also provides a context for applying statistical techniques. The dataset used in this project allows for the use of ANOVA, t-tests, Chi-square tests, logistic regression, and multiple linear regression to integrate programming, statistical methods, and research design. The motivation for commercial relevance but this project shows how to apply analytical and statistical techniques from data preprocessing to hypothesis testing and model evaluation.

The hope is to analyse the factors influencing transaction value both academically and practically. This allows for the development of analytical insights into customer behavior showing the significance of data cleaning, method selection and interpretation in data science contexts.

3 Literature Review

The studies based on consumer behavior have highlighted that transaction values are shaped by psychological, contextual, and product-specific factors. Different sources of literature have investigated how payment methods, product characteristics and time based patterns influence purchasing decisions and provide a foundation for analysing retail transaction data.

A significant area of research is based on the influence of payment method on consumer spending. Modern beahvoural economics suggests that digital and card-based transactions reduce the psychological "pain of paying" compared with cash leading to higher levels of spending. Work by (1) shows that credit card users are more willing to pay for very similar products with card which is one of the main explanations for increased trans-

action values associated with card payments. (2) found that mobile payment systems and how they reduce friction and save time increased purchase frequency while also decreasing perception of cost and changed purchase behavior. In (3) was also found that easy to use digital payment systems influence consumer decision-making improving convenience and easier purchases. These studies support the expectation that payment method influences transaction value making it an important factor in the current dataset.

Product category plays a key role in transaction values. Several studies show that spending patterns vary across product types. (4) analysed transaction-level data and found luxury and discretionary categories usually generate higher average transaction values. Categories such as groceries or household essentials tend to involve smaller and more consistent spending. This makes sense and is found in broader consumer behaviour literature which differentiates between necessity-driven and discretionary purchasing patterns.In (5) they highlight that dynamic pricing and promotional activity have stronger effects on discretionary spending categories due to flexibility in consumer decision-making. These studies both suggest that product categories make meaningful differences in transaction value consistent with the first research question.

Time based factors like differences between weekday and weekend behavior should also influence retail spending. (6) using large-scale bank account transaction data identified higher discretionary spending at weekends suggesting that consumers are more likely to engage in non-essential purchases during weekends. (7) showed that store atmosphere like background music affects shoppers differently across weekdays and weekends with weekend shoppers showing more impulsive and emotionally driven behaviors. This shows that time context may shape transaction value however these effects can differ across product categories.

Consumer behavior research has also explored how product complexity and perceived value influence spending decisions. In (8) they found that customers hesitate more when purchasing complex products or products they have to think about which can influence transaction values and purchasing patterns. The current dataset does not measure product complexity directly the distinction across product categories similar behavioral patterns may be found in different categories. Electronics or household goods may prompt higher spending compared with everyday consumables.

The reviewed literature supports the concept and basis of all four research questions in this project. Evidence suggests that payment method and product category influence spending behaviour and that time based patterns may produce measurable differences in transaction values. Transactional data can be used to model and predict consumer behaviour. The studies also show the importance of behavioral economics, psychological drivers, and influential factors in shaping transaction value all which justify the use of

statistical tests such as ANOVA, t-tests, Chi-square tests, and regression modelling in this analysis.

4 Data Description

The dataset used in this project contains 12,575 retail transactions collected from a multi-department retail store. Each row represents a single customer purchase and contains information describing the product purchased, the value of the transaction, the payment method used, and the date on which the transaction occurred. The dataset is structured in a tabular format and contains a mixture of numerical, categorical and time based variables making it suitable for statistical comparison and modelling. After preprocessing the final dataset kept its full size of 12,575 observations with missing or invalid entries addressed prior to analysis.

4.1 Dataset Structure

The dataset contains 13 variables, including three numerical variables, several categorical variables, and one datetime variable which additional features were taken. The main variables are:

- **total_spent (numeric):** The final transaction value, calculated after quantity and unit price.
- **price_per_unit (numeric):** The cost of a single unit of the purchased product.
- **quantity (numeric):** Number of units purchased in each transaction.
- **category (categorical):** The product category (e.g., Food, Beverages, Butchers, Furniture, Milk Products, etc.).
- **payment_method (categorical):** Method used to complete the transaction, such as cash, card, or digital wallet.
- **transaction_date (datetime):** The date and time the transaction occurred.
- **day_of_week (categorical):** Derived from the transaction date, indicating Monday–Sunday.
- **is_weekend (binary):** Indicator variable identifying whether the transaction took place on a Saturday or Sunday.

These variables form body of the research questions addressed in this report and support analyses comparing transaction values across categories, time periods, and payment types.

4.2 Initial Data Quality and Issues Identified

The dataset had several issues common to retail transaction logs. A number of categorical areas had inconsistent labelling (variations in category names and payment method formatting). Some datetime values required conversion from object format to Python datetime. Numerical fields like price and quantity were already in consistent formats but they were inspected for unrealistic values.

Some outliers in total-spent, price-per-unit, and quantity were identified visually. These showed genuine variation between product types (high-value categories such as Furniture vs lower-priced categories such as Milk Products). Because these values were sensible retail values they were kept.

There was little missing values found. In most cases missing indicators related to discount status or payment method were either filled with an appropriate category or left when statistically acceptable. No rows required removal.

4.3 Variables

Two time based variables were created to support the hypothesis tests:

- **day_of_week:** Used for visualising spending patterns across the week and included in regression modelling.
- **is_weekend:** Used to distinguish weekday vs weekend transactions for the t-test in RQ2.

These transformations ensured the dataset reflected both product attributes and behavioral context.

4.3.1 Summary

The dataset provides a good base for examining consumer spending patterns. The mix of numerical, categorical, and time based variables allows for a comprehensive analysis of the factors influencing transaction value. This supports the application of ANOVA, t-tests, Chi-square tests, logistic regression, and linear regression models in later sections.

5 Data Preprocessing

Data preprocessing was neccesary to ensure the dataset was suitable for statistical testing and modelling. The raw data contained a mixture of numerical, categorical, and temporal variables with several missing values and inconsistencies. A systematic cleaning process was conducted using Python.

Data Type Conversion

The `transaction_date` variable was initially stored as a string and was converted to a proper `datetime` format to allow time based feature extraction. From this cleaned date two additional variables were obtained:

- **day_of_week** – identifying the weekday name (Monday–Sunday)
- **is_weekend** – a binary variable for Saturday/Sunday transactions

These time based variables were required for hypothesis testing in RQ2 and for use in later regression modelling.

Handling Missing Values

Inspection of the dataset revealed missing values in several variables. Numerical variables such as `price_per_unit`, `quantity`, and `total_spent` contained missing entries. Because these variables are used to understand transaction value any rows missing these values were fixed. Categorical variables such as `item` and `discount_applied` also contained missing observations. Since these fields were not used as primary predictors in the statistical tests missing values were kept where appropriate or treated as non-applicable. This approach kept the full dataset size of 12,575 records.

Outlier Detection and Treatment

Outliers were examined in the three numerical variables: `price_per_unit`, `quantity`, and `total_spent`. Visual analysis and descriptive statistics showed a small number of large values in `total_spent`. These high values were consistent with purchases from higher-priced categories such as Furniture or Butchers and were treated as genuine transactions rather than errors. No extreme values were removed.

Standardisation of Categorical Variables

To ensure consistency in the analysis categorical fields were standardised. Differences in formatting ("Credit card" vs "Credit Card") were corrected and product categories were checked for duplicate labels coming from inconsistent capitalisation or spacing. Standardising these labels was neccesary for accurate group comparisons in ANOVA, the t-test, and the Chi-square test.

Final Clean Dataset

After completing all preprocessing steps the dataset remained at its full size of 12,575 rows and 13 variables. All numeric variables essential to the analysis contained no missing values, categorical variables were standardised and the time based feature provided additional behavioral context needed for the hypothesis tests. The cleaned dataset provided a reliable and well-structured basis for statistical analysis and modelling in each section.

6 Formulation of Hypotheses

The research questions influence the statistical methods used and determine the null and alternative hypotheses to be tested. Each hypothesis is stated and corresponds to the analytical technique applied in later sections.

Research Question 1 (RQ1)

Do average transaction values differ across product categories?

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

$$H_1 : \text{At least one product category has a different mean transaction value.}$$

This hypothesis is tested using a one-way ANOVA for comparing means across multiple independent groups.

Research Question 2 (RQ2)

Is there a significant difference in transaction value between weekday and weekend purchases?

$$H_0 : \mu_{\text{weekday}} = \mu_{\text{weekend}}$$

$$H_1 : \mu_{\text{weekday}} \neq \mu_{\text{weekend}}$$

An independent samples t-test is used to compare mean transaction values across these two time-based groups.

Research Question 3 (RQ3)

Is payment method associated with product category?

$$H_0 : \text{Payment method and product category are independent.}$$

$$H_1 : \text{Payment method and product category are dependent.}$$

This hypothesis is tested using a Chi-square test of independence based on a contingency table of categories and payment methods.

Research Question 4 (RQ4)

Can transaction value be predicted using categorical variables and key numeric attributes?

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$$

$$H_1 : \exists\, \beta_j \neq 0$$

Multiple linear regression is used to evaluate whether product category, payment method, discount status, day of week, price per unit, and quantity contribute significantly to predicting transaction value.

7 Methodology

This project uses quantitative research methodology combining data preprocessing, descriptive statistics, hypothesis testing, and predictive modelling. All analyses were conducted in Python using the `pandas`, `numpy`, `scipy`, `matplotlib`, `seaborn`, and `statsmodels` libraries. These tools helped with data manipulation, statistical testing and model estimation.

Analytical Process

The overall approach was as follows: (1) preparation and cleaning of the dataset, (2) form the hypotheses from the research questions, (3) application of appropriate statistical tests, and (4) interpretation of results using both statistical outputs and visualisation. The selection of methods ensures each test aligns with the type of variables involved and the structure of the research questions.

Descriptive Statistics and Visualisation

Initial descriptive statistics were produced to summarise the key variables including `total_spent`, `price_per_unit`, and `quantity`. Visualisations such as histograms, boxplots, and correlation heatmaps were generated to find normality, detect outliers, and explore relationships between variables. These descriptive analyses informed decisions about appropriate hypothesis tests and model specifications.

Method for RQ1: One-Way ANOVA

A one-way Analysis of Variance (ANOVA) was conducted. ANOVA is suitable for comparing means across more than two independent groups when the dependent variable is continuous. The model estimated was:

$$\text{total_spent} \sim \text{category}$$

Assumptions of ANOVA: normality of residuals and equal variances were assessed through residual diagnostics and visual inspection of boxplots. The F-statistic and p-value were used to determine whether differences in means across categories were statistically significant.

Method for RQ2: Independent Samples t-Test

The groups were formed from the binary variable `is_weekend`. Welch's t-test (`equal_var=False`) was used to account for potential differences in variances between the two groups. The test statistic and p-value were used to evaluate whether average spending differs across time periods. Visual comparison using boxplots helps with final results.

Method for RQ3: Chi-Square Test and Logistic Regression

A contingency table of payment method by category was constructed and the Chi-square statistic, degrees of freedom, and p-value were evaluated:

$$H_0 : \text{Payment method and category are independent}$$

To find patterns in digital payment a logistic regression model was estimated with a binary dependent variable to see if a transaction used a digital wallet. Predictors included product category and weekend status. Logistic regression was used because the outcome variable is binary. The model provided odds ratios and significance levels to assess the explanatory power of categorical factors.

Method for RQ4: Multiple Linear Regression

The initial model included only categorical predictors:

$$\text{total_spent} \sim C(\text{category}) + C(\text{payment_method}) + C(\text{day_week}) + \text{discount_applied}$$

Because this model explained little variance, an improved model was used including the two key numerical predictors:

$$\text{total_spent} \sim \text{price_per_unit} + \text{quantity} + C(\text{category}) + C(\text{payment_method}) + C(\text{day_week})$$

Regression diagnostics and adjusted R^2 were evaluated to compare model performance. This allowed for assessment of both behavioral and structural drivers of transaction value.

Together, these methods provide a good base for answering the research questions and supporting statistical conclusions.

8 Data Analysis

This section shows the results of the statistical analyses conducted to address the four research questions. The analysis uses descriptive statistics and correlation assessment using ANOVA, t-tests, Chi-square testing, logistic regression, and multiple linear regression modelling.

Descriptive Statistics

Descriptive statistics were produced to summarise the numerical variables `total_spent`, `price_per_unit`, and `quantity`. After preprocessing the results were:

- **Price Per Unit**: mean = 23.37, std = 10.48, median = 23.37, range = 5–41.
- **Quantity**: mean = 5.54, std = 2.79, median = 5.54, range = 1–10.
- **Total Spent**: mean = 128.24, std = 90.39, median = 112.50, range = 5–395.

Plots showed skewness, the presence of higher-value transactions in categories such as Furniture and Butchers, while categories such as Food and Milk Products showed narrower price ranges.

Correlation Analysis

A correlation matrix was created to examine linear associations in numerical variables. The key findings were:

- **total_spent** and **quantity**: $r = 0.7045$ (strong positive)
- **total_spent** and **price_per_unit**: $r = 0.5882$ (moderate positive)
- **price_per_unit** and **quantity**: $r = 0.0023$ (near zero)

These results confirm that the strongest drivers of transaction value are the units purchased and their unit cost. The variable `is_weekend` had correlations close to zero with all numeric variables showing quite small differences between weekday and weekend spending patterns.

RQ1: One-Way ANOVA – Category Differences

$$\text{total_spent} \sim \text{category}$$

The results were:

- **F-statistic**: 5.10
- **p-value**: 0.000008

Since the p-value is far below 0.05 the null hypothesis is rejected. There are statistically significant differences in average transaction value across categories.

Boxplots showed that categories such as Butchers, Beverages, and Furniture have higher medians, while Milk Products and Food show lower spending. This reflects price variation across product groups.

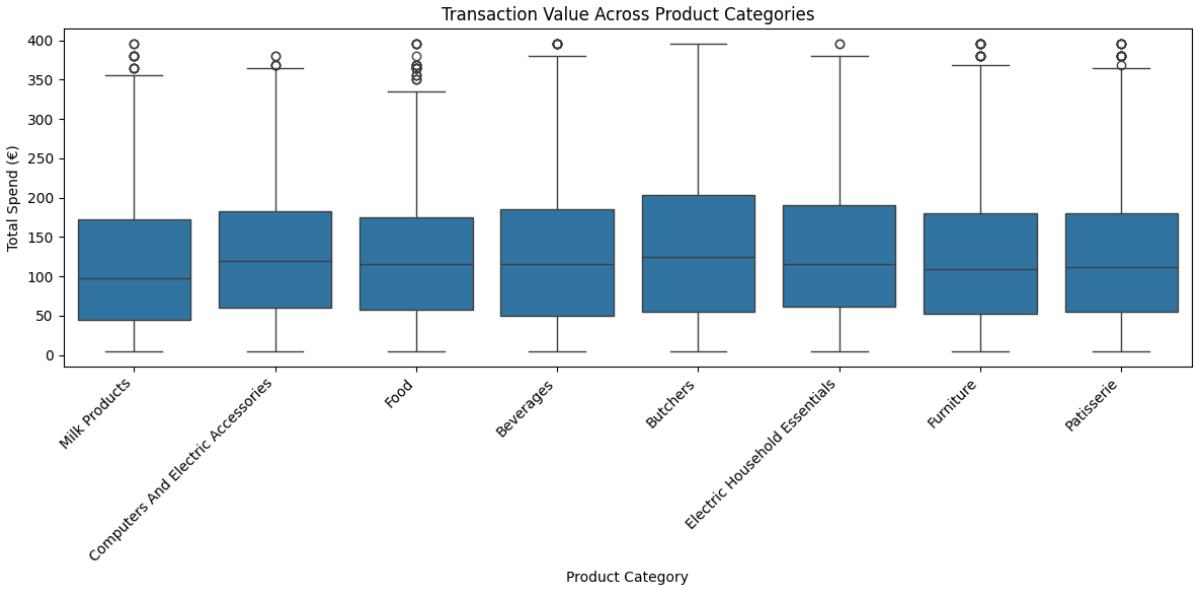

Figure 1: Transaction Value Across Product Categories

RQ2: Independent Samples t-Test – Weekday vs Weekend

The results were:

- **t-statistic**: -0.7750
- **p-value**: 0.4383

Because the p-value exceeds 0.05 the null hypothesis cannot be rejected. Average spending is not significantly different between weekday and weekend transactions.

Although visual plots showed slight increases in weekend spending for some categories (Beverages, Butchers), these were not able to produce statistical significance in the overall test.

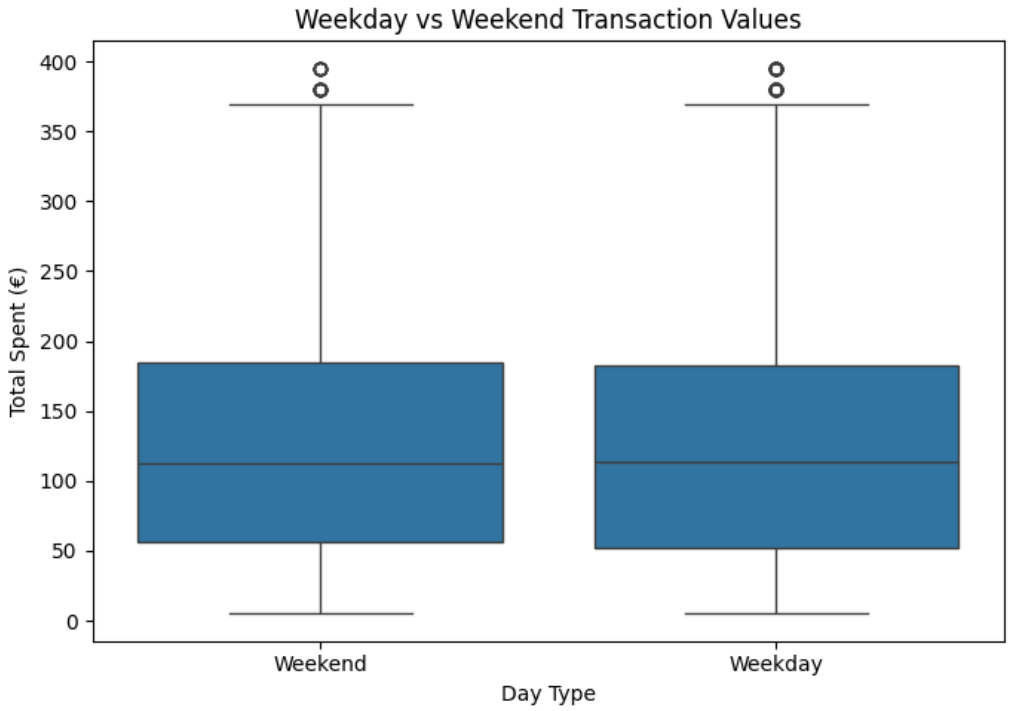

Figure 2: Weekday vs Weekend Transaction Values

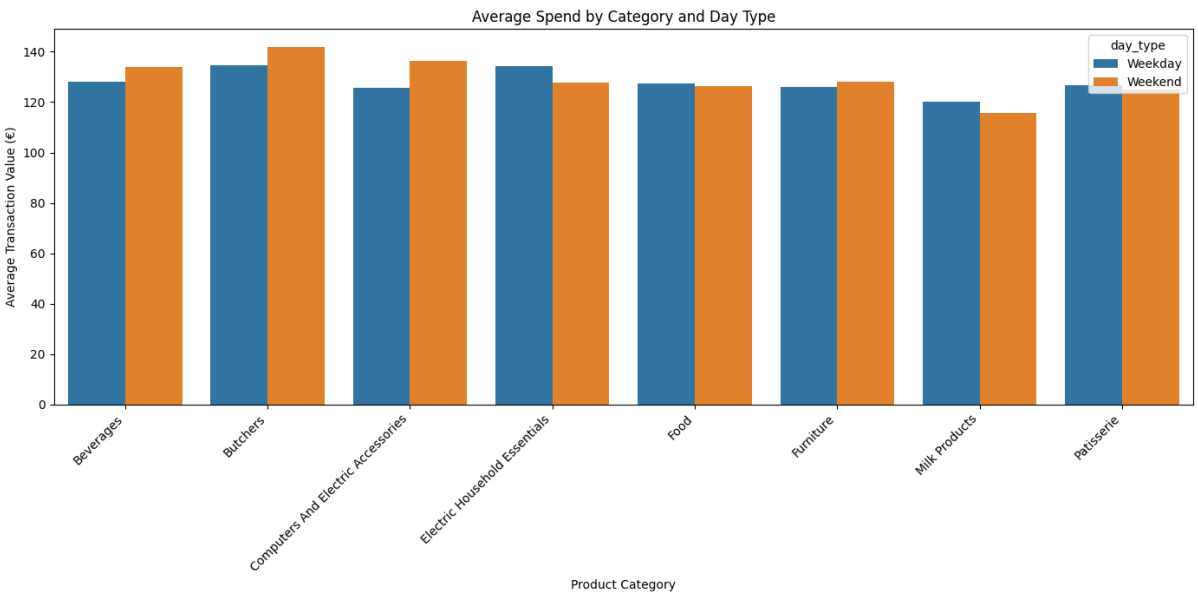

Figure 3: Average Spend by Category and Day Type for better visualisation

RQ3: Chi-Square Test – Payment Method and Category

The results were:

- **Chi-square statistic**: 21.01

- **df**: 14
- **p-value**: 0.101

Since the p-value is greater than 0.05 payment method and product category are not significant. Usage of cash, card, and digital wallets appears consistent across categories.

Logistic Regression – Predicting Digital Wallet Use

To further examine payment method behaviour a logistic regression was estimated:

$$\text{wallet_user} \sim C(\text{category}) + \text{is_weekend}$$

Results included:

- **Pseudo** $R^2 = 0.0005$
- **Overall model p-value** $= 0.4006$

No predictors were significant and all odds ratios were close to 1. This supports the Chi-square finding that payment method usage is not influenced by product type or time of week.

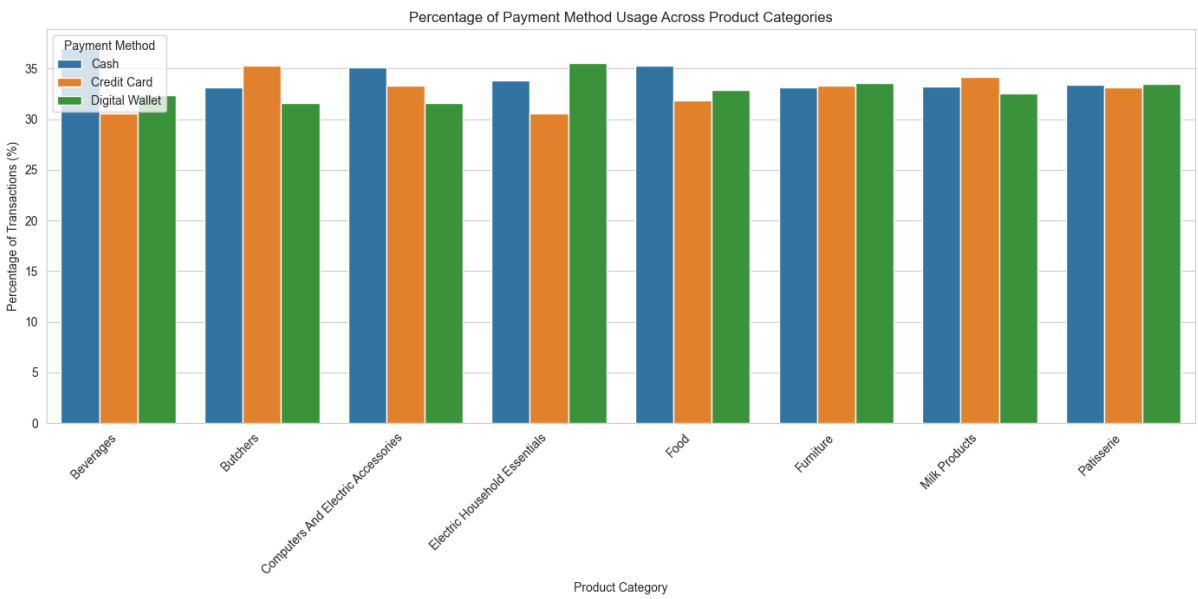

Figure 4: Percentage of Payment Method Usage Across Product Categories

RQ4: Multiple Linear Regression – Predicting Transaction Value

Two regression models were constructed.

Model 1: Categorical Predictors Only

$$\text{total_spent} \sim C(\text{category}) + C(\text{payment_method}) + C(\text{day_of_week}) + \text{discount_applied}$$

Results:

- $\mathbf{F(15,\ 12{,}559)} = 3.357$
- $\mathbf{p} = 1.09 \times 10^{-5}$
- $\mathbf{R}^2 = 0.004$

The model is statistically significant but explains less than 1% of transaction value showing that categorical factors alone have limited predictive power.

Model 2: Adding Price and Quantity

A second model included the two key numeric drivers:

$$\text{total_spent} \sim \text{price_per_unit} + \text{quantity} + C(\text{category}) + C(\text{payment_method}) + C(\text{day_of_week})$$

Results:

- $\mathbf{R}^2 = 0.833$

Once the main drivers of the transaction were included model performance improved greatly. Price per unit and quantity were highly significant while category and payment method lost significance showing that most variation in transaction value is explained by these two numerical predictors.

Summary of Findings

The findings of the analysis can be summarised as follows:

- Product category significantly affects transaction value.
- Weekday vs weekend spending shows no significant difference.
- Payment method usage does not vary meaningfully across categories.
- Price per unit and quantity are the dominant predictors of transaction value.

9 Interpretation

The results allow us to understand the factors that influence transaction value within the retail dataset. The ANOVA test for RQ1 showed that average spending differs significantly across product categories confirming that some types of goods generate higher-value purchases. Categories such as Butchers, Beverages, and Furniture were associated with higher transaction values while Food and Milk Products showed lower spending. This shows structural differences in product pricing and purchase quantities play a major role in shaping customer spending behavior.

The independent samples t-test for RQ2 showed no statistically significant difference between weekday and weekend spending. Slight visual differences appeared for certain categories these variations were not enough to produce significance at the overall level. This suggests that customers spend similar amounts regardless of when they shop and that time based factors do not influence transaction value in this dataset.

The Chi-square test for RQ3 showed that payment method is not significantly associated with product category. This finding was supported by the logistic regression model which showed that neither product category or weekend status meaningfully predicted digital wallet usage. This shows that payment choices do not vary by product type or shopping time.

In RQ4 The initial regression model using only categorical predictors had extremely low explanatory power showing that variables such as product category, payment method, and day of the week explain very little variation in spending. Once the key numerical predictors—price per unit and quantity—were included in the improved regression model the explanatory power increased. This confirms that transaction value is driven by the structure of the purchase itself rather than behavioral or time based factors. Price and quantity account for majority of variation in spending.

These findings show a consistent theme. Customer spending is determined more by the characteristics of the products purchased than by when customers shop or how they pay. Behavioral differences exist at the category level but they are outperformed by the main drivers in the transactional data.

10 Conclusion

This project set out to identify the factors that most strongly influence transaction value within a large retail dataset and to assess whether spending behavior varies across product categories, time periods, and payment methods. The results clearly show that product category is the only categorical variable with a meaningful effect on transaction value.

Higher-priced categories such as Butchers, Beverages, and Furniture produce larger transactions, while other categories have lower spending patterns.

Time based patterns had little impact on spending. The t-test confirmed that weekday and weekend transactions do not differ significantly in value suggesting that customers have similar spending habits during the week. Payment method usage was consistent across product categories and logistic regression showed that neither category nor weekend status predicts digital wallet usage.

The regression analysis provided the strongest insight. Categorical factors alone explained almost no variance in spending but once price per unit and quantity were included model performance increased. This confirms that the main influences of transaction value are found directly in the structure of the purchase.

Overall the findings show that retail spending is shaped far more by what is bought and how much of it than when or how the purchase is made. These results highlight the importance of product level characteristics in retail analytics and highlight the value of statistical modelling.

11 Future Works

While this project provides good insights into the influences of transaction value there are ways to improve the analysis. Future work could use other customer level features such as demographics, loyalty membership, or purchasing history. These variables could show behavioral patterns that are not captured in the current dataset.

Second analysing full shopping baskets rather than single item transactions would show a more realistic view of consumer behavior. Basket size, product combinations, and cross category spending could provide better insights into purchasing decisions.

Third introducing time-series analysis such as seasonality, monthly trends, or promotional periods could help show longer-term patterns. This would be useful for forecasting and inventory planning.

These methods would build on the foundations of this analysis and support better retail insights.

12 Reflective Journal

Completing this project allowed me to experience a full data analysis workflow in a practical way. One of the main lessons I learned was the importance of data preprocessing.

Although the dataset appeared straightforward initially after examining missing values, converting date formats, and standardising categorical labels showed me how important it is to clean data carefully before any modelling can take place. I realised that small inconsistencies can create misleading results later in the analysis.

Another area of learning was choosing appropriate statistical methods. Applying ANOVA, t-tests, Chi-square tests, logistic regression, and multiple linear regression helped me understand how these techniques work and also why certain tests are suitable for different types of variables and research questions. The comparison between the initial regression model and the improved model highlighted the difference between statistical significance and explanatory power, teaching me results should always be interpreted in context.

Visualisation played a major role in my understanding. Heatmaps, boxplots, and grouped bar charts revealed patterns that were not immediately visible from numerical summaries alone. These visuals made it easier to communicate findings clearly and supported my interpretation of the statistical results.

Overall this project helped strengthen my analytical thinking and improve my confidence in combining coding, statistical reasoning, and research skills. It showed me how the different components of the data analytics process connect from forming hypotheses to interpreting models and provided a sense of how data-driven insights are generated in practice.

Appendix A: Data Preparation Summary

The following data-cleaning steps were applied prior to analysis (full code submitted separately):

- Replacement of missing and nonsense values.
- Standardisation of categorical labels.
- Conversion of dates and creation of time-based variables.
- Removal of duplicates and invalid numeric entries.
- Outlier detection using the IQR method with imputation.
- Recalculation of total transaction values.

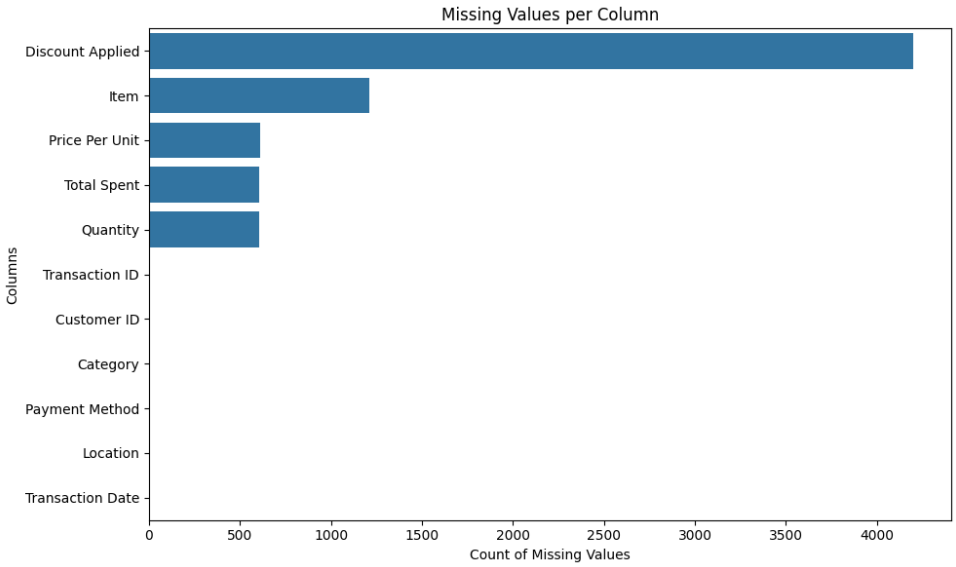

Figure 5: Missing values per column before data cleaning.

Appendix B: Exploratory Data Analysis Figures

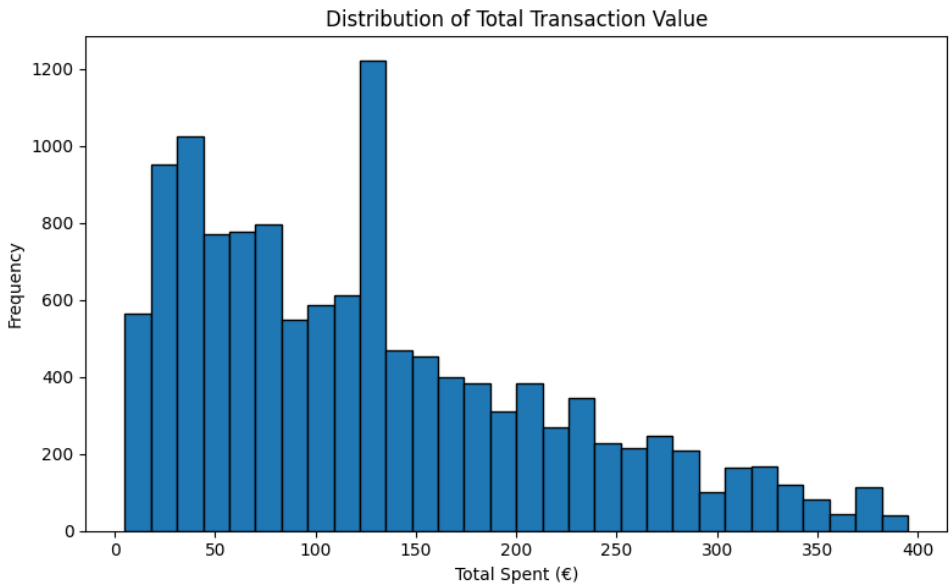

Figure 6: Histogram of total transaction value.

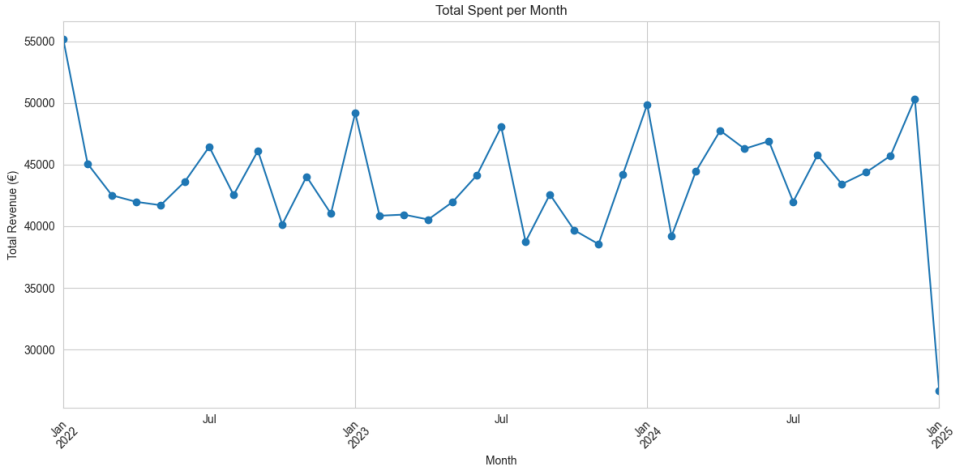

Figure 7: Total spending across product categories.

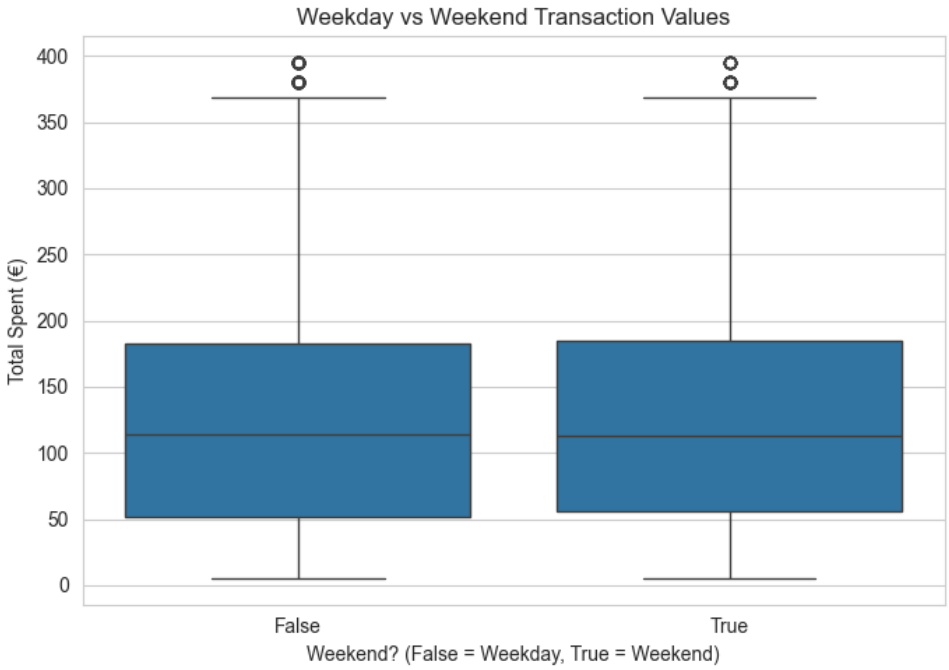

Figure 8: Weekday versus weekend spending.

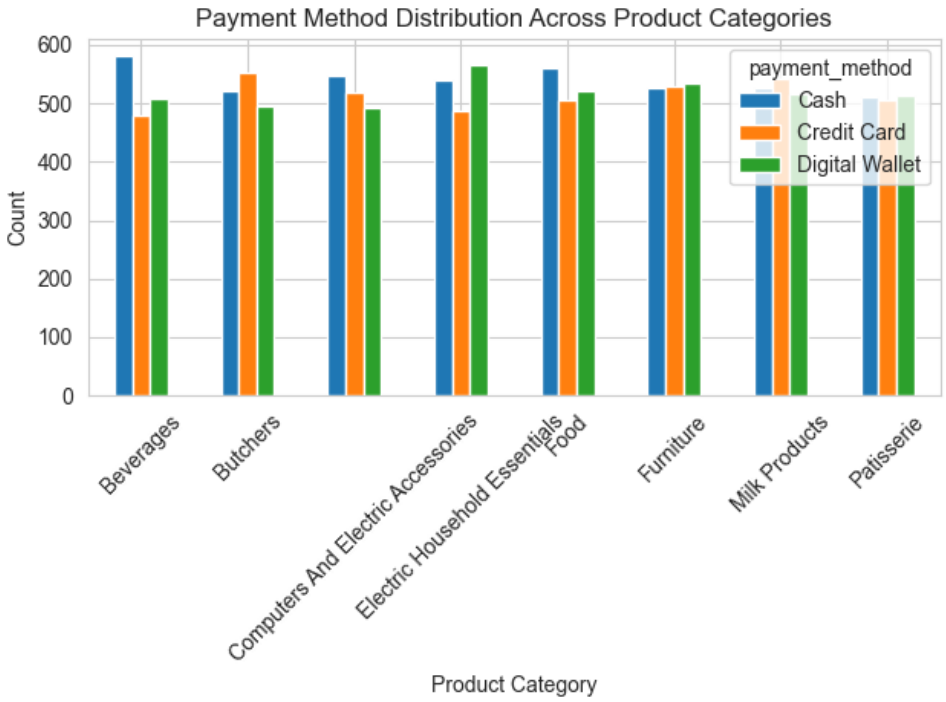

Figure 9: Spending patterns across categories for weekdays and weekends.

Appendix C: Statistical Test Outputs

C1. ANOVA (RQ1)

ANOVA summary table and diagnostics:

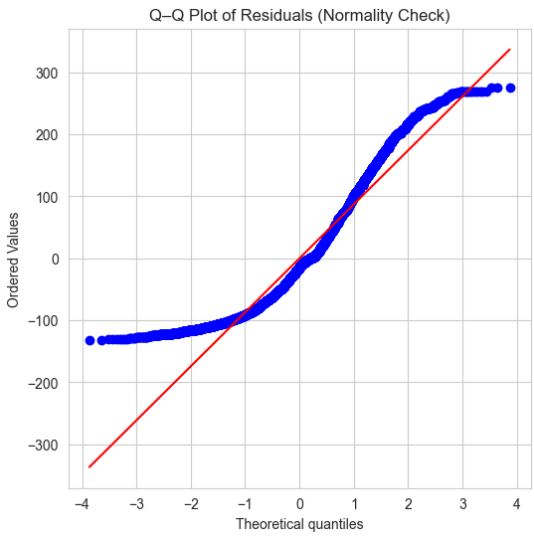

Figure 10: Q–Q plot of ANOVA residuals.

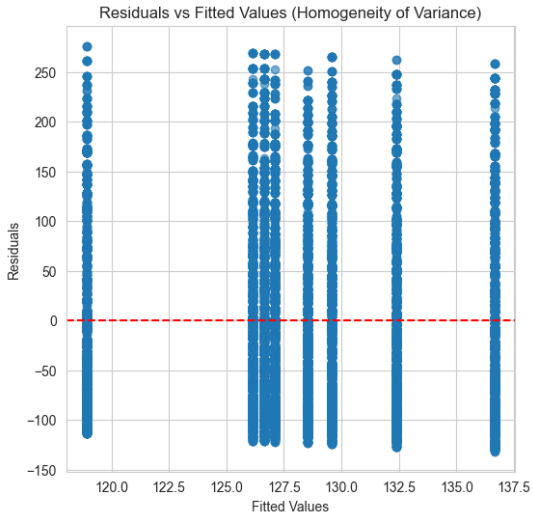

Figure 11: Residuals vs fitted plot for ANOVA model.

C2. Independent Samples t-Test (RQ2)

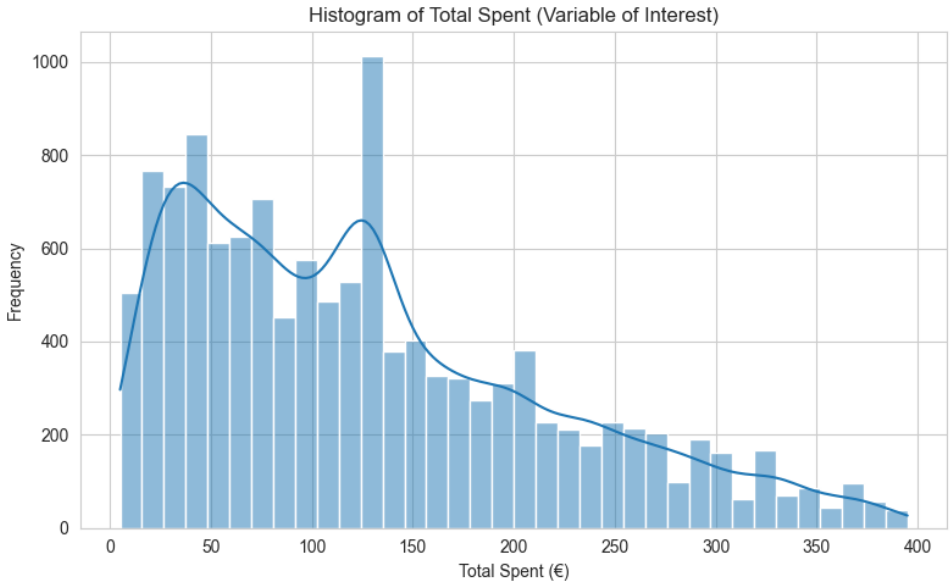

Figure 12: Distribution of total spending for weekdays and weekends.

C3. Chi-Square Test (RQ3)

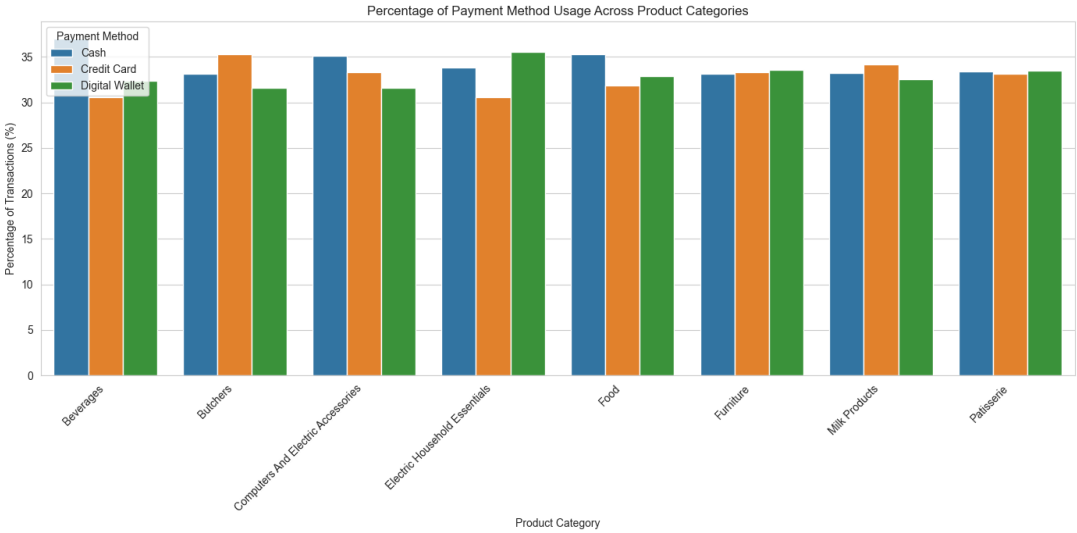

Figure 13: Payment method distribution by product category.

Appendix D: Regression Diagnostics (RQ4)

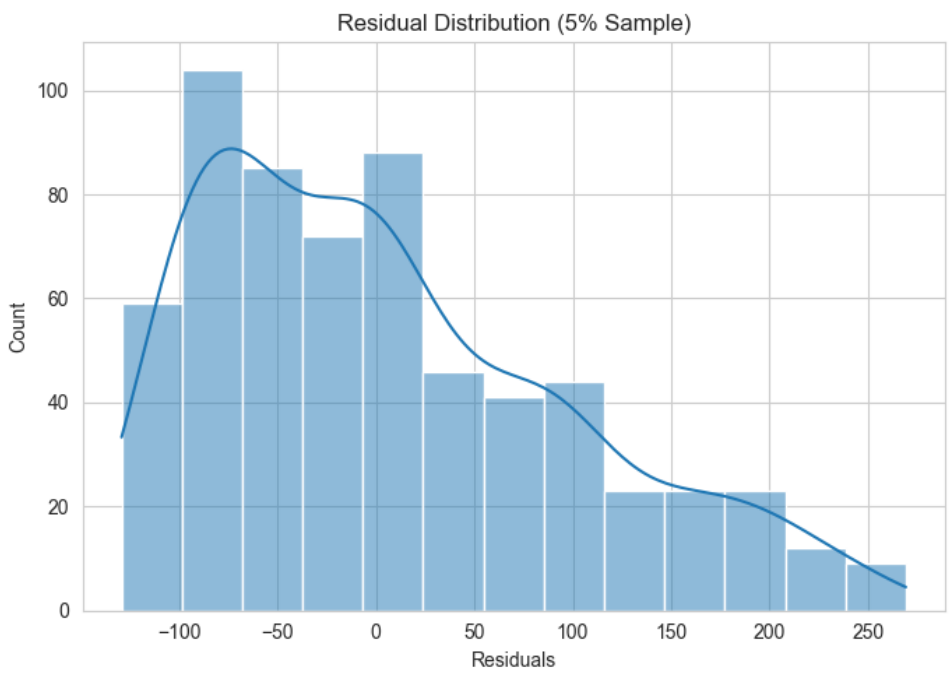

Figure 14: Residual distribution (5% sample) for the regression model.

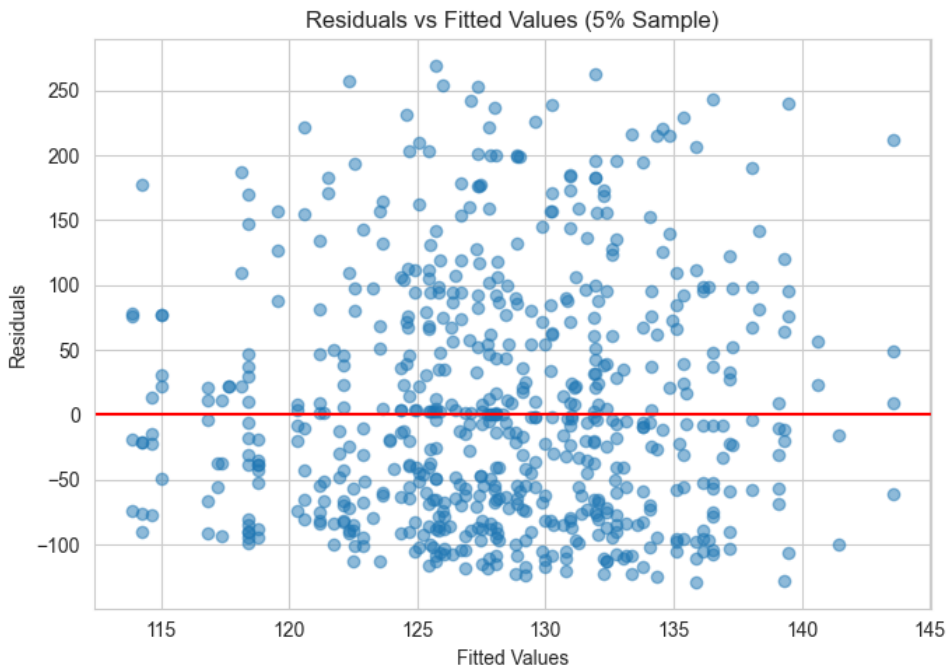

Figure 15: Residuals vs fitted values (5% sample).

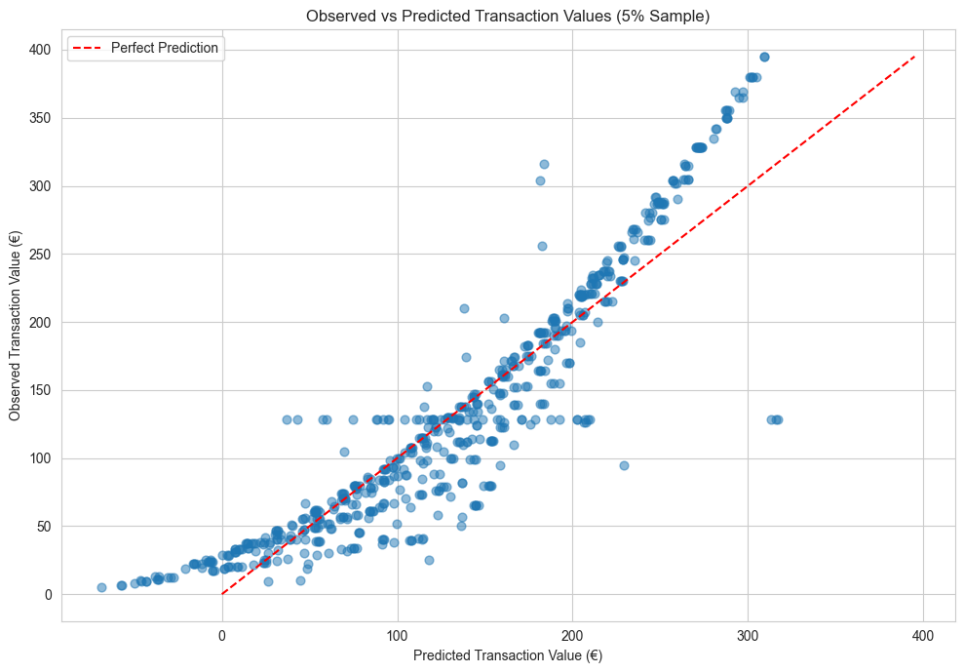

Figure 16: Observed vs predicted total spending (5% sample).

Appendix E: Code Files (Submitted Separately)

All data-cleaning and statistical analysis code is submitted separately as required:

- `Final_Project_Cleaning.py`

- `Final_Project_DataAnalysis.py`
- HTML notebook exports:
  - `Final_Project_NewDataSet.html`
  - `Final_Project_DataAnalysis.html`

These contain the full implementation of all methods, visualisations, and diagnostics used in the report.

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